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THE STATE OF THE FIELD

Using differential introgression in hybrid zones to identify genomic regions involved in speciation

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Abstract

Hybrids between species provide information about the evolutionary processes involved in divergence. In addition to creating hybrids in the laboratory, biologists can take advantage of natural hybrid zones to understand the factors that shape gene flow between divergent lineages. In the early stages of speciation, most regions of the genome continue to flow freely between populations. Alternatively, the subset of the genome that confers reproductive barriers between nascent species is expected to reject introgression. Now enabled by advances in genomics, this perspective is motivating detailed comparisons of gene flow across genomic regions in hybrid zones. Here, I review methods for measuring and interpreting introgression at multiple loci in hybrid zones, focusing on the problem of identifying loci that contribute to reproductive isolation. Emerging patterns from multi-locus studies of hybrid zones are highlighted, including remarkable variance in introgression across the genome. Although existing methods have been useful, there is scope for development of new analytical approaches that better connect differential patterns of gene flow in hybrid zones with current knowledge of speciation mechanisms. I outline future prospects for differential introgression studies on a genomic scale.

Keywords: gene flow, hybrid zones, hybridization, introgression, secondary contact, speciation

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Hybrid zones, speciation and introgression

Hybridization between species is common in nature (Arnold 1997, 2006; Rieseberg 1997; Mallet 2005). One consequence of this mixing is the existence of regions of secondary contact that grade from one species to the other over relatively short geographic distances. In principle, these hybrid zones can be maintained by increased fitness of hybrids along narrow ecotones (Moore 1977) or an equilibrium between migration and selection acting along an environmental gradient (Fisher 1930; Haldane 1948; Endler 1977). However, the observations that changes in diverse phenotypes are usually coincident, regions of hybridization are geographically narrow, and natural hybrids often suffer reduced fitness (Barton & Hewitt 1985, 1989; Burke & Arnold 2001) suggest that most hybrid zones reflect a balance between migration and selection against hybrids (Key 1968; Bazykin 1969; Barton & Hewitt 1989). The reduced fitness of hybrids and its consequences for gene exchange between species connect the dynamics of hybrid zones to the genetics of speciation.

Hybrid zones can be viewed as natural arenas in which combinations of mutations that have accumulated between species are functionally tested. These mutations can be loosely sorted into three categories. One group of variants is equally fit in hybrids and pure species. These mutations are largely free to flow across hybrid zones. A second subset of variants thrives on hybrid backgrounds, moving quickly across regions of secondary contact. A third collection of mutations disrupts reproduction or development in hybrids (postzygotic isolation) or engenders conspecific mating preferences (prezygotic isolation), inhibiting gene flow between species. Hybrid genomes are complex combinations of these three types of variants, mixed together by migration and recombination.

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Decades of hybrid zone surveys using allozymes and DNA markers have demonstrated that different genomic regions are exchanged between species at different rates (Hunt & Selander 1973; Dowling *et al.* 1989; Harrison 1990). Recent studies employing larger numbers of markers confirm that this inter-locus variation is often substantial (Table 1). This pattern of differential introgression indicates that species boundaries are 'semipermeable' (Barton & Hewitt 1981; Harrison 1986, 1990; Wu 2001): a subset of loci confers reproductive barriers, maintaining species integrity in the face of gene flow. Hybrid zones act as selective filters that 'catch' genomic regions responsible for reproductive isolation while allowing other pieces of the genome to pass (Barton & Hewitt 1985). As species diverge, more loci participate in reproductive barriers and the fraction of the genome that remains exchangeable between species declines.

Knowing the specific genes and mutations that underlie reproductive isolation between species provides access to the evolutionary and molecular mechanisms that drive speciation, including the role of natural selection (Orr *et al.* 2004). Motivation for identifying these loci has recently surged as a result of several success stories in a handful of taxa (Wittbrodt *et al.* 1989; Ting *et al.* 1998; Barbash *et al.* 2003; Presgraves *et al.* 2003; Brideau *et al.* 2006; Masly *et al.* 2006; Mihola *et al.* 2009; Phadnis & Orr 2009). Detailed descriptions of gene flow across natural hybrid zones enrich the picture of speciation obtained from genetic studies of reproductive isolation in the laboratory. Whereas laboratory crosses attribute specific hybrid phenotypes to particular loci, patterns of molecular variation in hybrid zones directly measure the capacity of species to exchange genes when their ranges overlap. Hybrid zones thus provide powerful natural forums for evaluating the effects of known reproductive isolation loci and for identifying new genetic barriers to hybridization.

Here, I review methods for measuring and interpreting introgression at multiple loci, in the context of speciation. I focus on spatial and genomic analyses applied to hybrid zones, ignoring other useful approaches that focus on genetic differentiation among a few sympatric or allopatric populations (Turner *et al.* 2005; Yatabe *et al.* 2007; Fitzpatrick *et al.* 2009; Kane *et al.* 2009). Surveys of differential gene flow have the potential to localize genomic regions that introgress adaptively between species, a phenomenon for which there is growing empirical evidence (Arnold 1997, 2006; Kim & Rieseberg 1999; Rieseberg *et al.* 1999; Payseur *et al.* 2004; Martin *et al.* 2006; Whitney *et al.* 2006; Teeter *et al.* 2008; Arnold & Martin 2009; Fitzpatrick *et al.* 2009). Although this review concentrates on the challenge of identifying loci that contribute to reproductive isolation, similar logic applies to the detection of adaptive introgression as well.

Measuring and comparing introgression at individual loci in hybrid zones

The following description of methods for analysing introgression in hybrid zones focuses on the common scenario in which two lineages have accumulated genetic differences and reproductive barriers (in the form of prezygotic and/or postzygotic isolation) in allopatry and subsequently come into secondary contact (Mayr 1963). I assume that individuals comprising a series of populations have been sampled across a spatial transect of the hybrid zone and genotyped at a series of molecular markers. All markers have two alleles in the sample, as is typical for single nucleotide polymorphisms (SNPs). Methods exist for transforming variation at highly polymorphic loci (e.g. microsatellites) to two allele systems for the purposes of measuring introgression (Gregorius & Roberds 1986).

Geographic clines

Most models used to analyse hybrid zones describe the relationship between allele frequency and geography. In this framework, allele frequencies measured at a series of localities are compared to the geographic distances of these localities along a hybrid zone. Examples of allele frequency clines at 10 loci in a *Passerina* bunting hybrid zone (Carling & Brumfield 2009) are displayed in Fig. 1. Hybrid zone clines often assume this sigmoidal shape, with a sharper transition near the centre and shallower decays towards the edges. The step in allele frequency arises at the initiation of secondary contact, even without selection acting at the locus in question. Over time, neutral diffusion flattens the cline, at a rate proportional to the square root of the number of generations since contact (Endler 1977, p. 91). Hence, selection is required to maintain the sigmoidal form. Either different environments must favour alternative alleles, leaving hybrids less adapted to their environment, or combinations of alleles (within or between loci) must intrinsically reduce hybrid fitness wherever they occur. Under both scenarios, allele frequency clines are primarily shaped by the interaction between selection and migration. In particular, the spatial change in allele frequency is expected to occur on a characteristic scale (w), defined by

$$w = k\frac{\sigma}{\sqrt{s}} \quad (1)$$

where σ is the standard deviation of the distance between parents and offspring and s is the strength of selection. This relationship holds for a variety of migration-selection models of single loci, with k taking on different values under contrasting selection regimes (Slatkin 1973; Nagylaki 1975; Endler 1977; Barton & Gale 1993). For example, $k = 2$ when selection acts against heterozygotes

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Table 1 Studies of differential introgression across hybrid zones. Only studies specifically designed to measure differential introgression among at least 10 DNA markers in hybrid zones are included. Organization is alphabetical by species pair

Species pair	Organism	Surveyed markers	Analytical approach	Major results	Reference
Cottus perifretum – *Cottus rhenanus*	Sculpin	168 microsatellites	Genomic clines	Deviation from neutral model at 85 loci; many loci show selection against heterozygotes or directional selection; limited correspondence of same markers in two transects	Nolte *et al.* 2009
Helianthus annuus – *Helianthus petiolaris*	Sunflower	88 RAPDs	Genomic clines	Reduced introgression near QTL for pollen sterility and on rearranged chromosomes	Rieseberg *et al.* 1999
Helianthus annuus – *Helianthus petiolaris*	Sunflower	61 RAPDs	Genomic clines	Congruent introgression in geographically distant hybrid zones; non-neutral introgression at many loci	Buerkle & Rieseberg 2001; reanalysed by Gompert & Buerkle 2009
Mus domesticus – *Mus musculus*	House mouse	13 X-linked SNPs	Geographic clines; genomic clines; LD	Reduced introgression in the centre of the X chromosome	Payseur *et al.* 2004; reanalysed by Gompert & Buerkle 2009
Mus domesticus – *M. musculus*	House mouse	7 autosomal allozymes; 3 X-linked SINEs, 2 X-linked SNPs	Geographic clines	Reduced introgression at X-linked loci relative to autosomal loci	Macholan *et al.* 2007
Mus domesticus – *M. musculus*	House mouse	39 autosomal SNPs; 13 X-linked SNPs	Geographic clines; LD	50-fold variation in introgression; LD among conspecific alleles on different chromosomes; reduced introgression of X-linked loci	Teeter *et al.* 2008
Mus domesticus – *M. musculus*	House mouse	38 autosomal SNPs; 3 X-linked SNPs	Geographic clines; genomic clines; LD	Substantial variance in introgression; LD among conspecific alleles on different chromosomes; limited correspondence of same loci in two transects	Teeter *et al.* 2010
Passerina amoena – *Passerina cyanea*	Bunting	10 resequenced Z-linked loci	Geographic clines	Cline width similar for most loci; introgression strongly reduced at 1 locus, which may be involved in reproductive isolation	Carling & Brumfield 2009
Populus alba – *Populus tremula*	Poplar	19 microsatellites	Genomic clines	*P. tremula* alleles over-represented at multiple loci and under-represented at 1 locus	Lexer *et al.* 2007

SNPs, single nucleotide polymorphisms.

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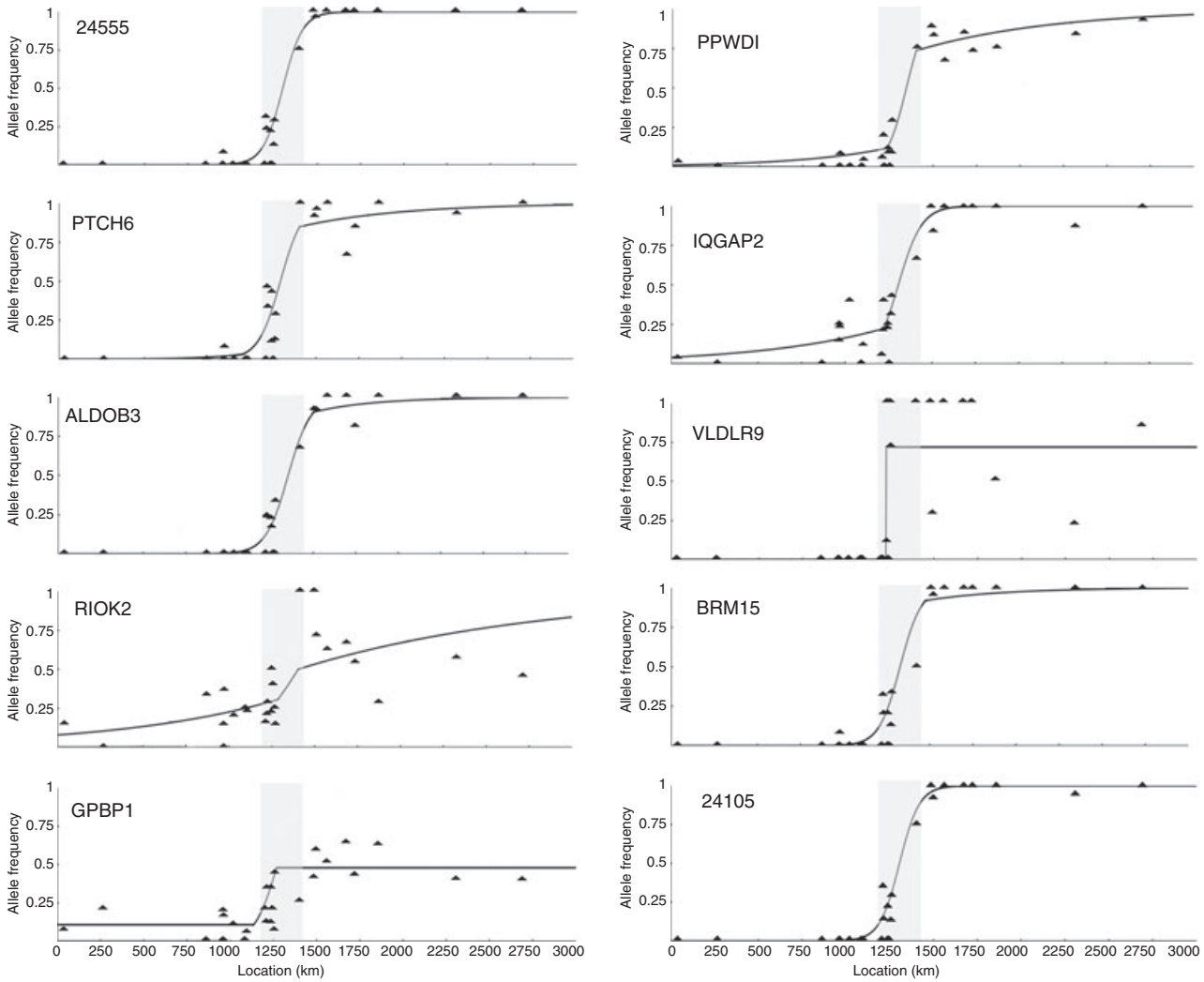


Fig. 1 Allele frequency clines from 10 Z-linked loci in a North American hybrid zone between *Passerina amoena* and *Passerina cyanea*. Each panel shows data from one locus, each triangle represents a locality, and lines depict clines fitted by maximum likelihood. Locus name is indicated in the top left corner of each panel. Grey boxes show approximate geographic location of the hybrid zone. Reproduced from Fig. 2 in Carling & Brumfield (2009).

and $k = 1.782$ when selection favours different alleles on the two sides of the hybrid zone (Barton & Gale 1993). This equation demonstrates a central principle governing genetic variation in hybrid zones: dispersal tends to widen the gradient of allele frequency, while selection tends to sharpen the gradient. Because the characteristic scale is measurable, clines at individual loci provide insight into this balance between migration and selection.

w is also referred to as cline width and can be estimated (in geographic units) as the inverse of the maximum slope of the relationship between allele frequency and geographic position. Another useful variable is the geographic point at which the maximum allele frequency gradient is observed. This point often lies in the middle of hybrid zones and is called cline centre (c).

The sigmoidal shape of allele frequency clines approximately translates into three straight lines when allele frequency (p) is plotted on a logit scale. In this case, heterozygote disadvantage produces a linear cline in the centre of the hybrid zone,

$$\ln\left(\frac{p}{1-p}\right) = \frac{4x}{w} \qquad (2)$$

(Bazykin 1969), where x is the geographic position of a locality, and clines for other types of selection are also approximately linear on this scale (Barton & Gale 1993). Exponential increases and decreases in allele frequency on the edges of the hybrid zone also lead to straight lines of the form

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$$\ln\left(\frac{p}{1-p}\right) = \lambda x \tag{3}$$

where λ describes the exponential rate of allele frequency change (Szymura & Barton 1986, 1991).

Taking advantage of these relationships, Barton and colleagues developed a powerful analytical approach to estimate cline parameters in three segments of the hybrid zone (Szymura & Barton 1986, 1991; Porter *et al.* 1997). First, the relationship between allele frequency and distance along the transect (x) in the centre of the zone is approximated by

$$p = \frac{1}{2}\left[1 + \tanh\left(\frac{2[x-c]}{w}\right)\right] \tag{4}$$

The tanh function provides the characteristic sigmoidal shape. This model features two parameters, cline centre and cline width. Moving away from the centre of the zone, allele frequency decays exponentially according to

$$p = \frac{\exp\left[4(x - [c + z_1])\sqrt{\theta_1}\right]}{w} \tag{5}$$

on the side of the zone where species 1 resides, and according to

$$p = 1 - \frac{\exp\left[4(x - [c + z_2])\sqrt{\theta_2}\right]}{w} \tag{6}$$

on the side of the zone where species 2 lives. In addition to cline width, cline centre and geographic distance, these equations contain θ, the rate of exponential decay on each side, and z, the distance from the cline centre of vertical asymptotes in the exponential decay of allele frequencies. As θ approaches 1 and z approaches 0, clines from the edges of the hybrid zone resemble the cline in the centre of the zone. θ is generally <1, reflecting the slower decay of allele frequency on the edges of the zone relative to the sharp change in the centre.

Whereas introgression across the centre of the zone primarily reflects overall selection against hybrids (see Linkage disequilibrium, below), allele frequency dynamics at the edges of the zone are mostly influenced by locus-specific selection (Barton 1979; Barton & Bengtsson 1986). As a result, the strength of the reproductive barrier (B) experienced by a neutral locus also can be estimated by comparing the gradients in allele frequency in the centre and the edges (Nagylaki 1976):

$$B = \frac{\Delta p_{\text{center}}}{\left(\frac{\partial p}{\partial x}\right)_{\text{edge}}} \tag{7}$$

The numerator of this equation is the step in allele frequency across the middle of the zone, and the denominator is the decay of allele frequency on one edge of the zone. Asymmetries in migration or selection will produce distinct barriers on different sides of the zone. B can be viewed as the geographic distance between populations that would be needed to produce the difference in allele frequency observed in the centre of the zone under a neutral scenario. This equation also describes cline shape at the points where the steep central cline meets the exponentially decaying allele frequency.

Parameter estimation involves joint analysis of the three portions of the cline. Because simultaneous estimation of the six parameters is analytically intractable, numerical methods are used. Parameters are estimated by maximum likelihood. The log-likelihood of the observed set of allele frequencies in *individuals* (P), assuming a particular set of parameter values ($V = w, c, \theta_1, \theta_2, z_1, z_2$), is derived from the binomial distribution:

$$L(P, V) = \sum_i \sum_j \left[\sum_k p_{ijk} \ln \frac{p_{ijk}}{p[V, x_i]} + \sum_k (1 - p_{ijk}) \ln \frac{(1 - p_{ijk})}{(1 - p[V, x_i])}\right] \tag{8}$$

where p_{ijk} is the allele frequency in the jth individual at the kth locus in the ith population and $p(V, x_i)$ is the expected allele frequency at point x_i predicted from the equations above with parameter values V (Porter *et al.* 1997). Maximum likelihood estimates and support limits can be found by randomly sampling parameter space using Metropolis algorithms (Szymura & Barton 1986, 1991; Porter *et al.* 1997) or other approaches. Hypothesis tests can then be conducted using likelihood ratios.

When multiple loci are surveyed, this analysis can be used to separately estimate cline shapes for each locus. Because locus-specific selection is expected to generate inter-locus heterogeneity in introgression, the statistical comparison of these cline shapes is of great interest. The likelihood framework provides several solutions. Loci with non-overlapping two-log-likelihood support limits for parameter estimates might be judged to have significantly different cline shapes. A more powerful strategy is to re-estimate the likelihood of observing the data at one locus assuming the parameter values estimated for a different locus. Then, a likelihood ratio test evaluates the null hypothesis that the same parameter values describe cline shapes at the two loci. This comparison can be conducted by assuming that all parameters are shared between loci or by evaluating one parameter at a time. The likelihood ratio test has degrees of freedom equal to the number of fixed parameters in the reduced model. A similar test is available for asking whether significant heterogeneity exists when all loci are considered simultaneously (Phillips *et al.* 2004; Macholan *et al.* 2007). In this approach, likelihood profiles are summed across loci, and the maximum likelihood of the resulting composite profile is recorded. Next, maximum likelihoods are estimated for each locus separately and summed.

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If there is significant variation in introgression, the second likelihood will be higher than the first, a possibility again evaluated with a likelihood ratio test (with degrees of freedom equal to the number of loci minus one).

Heuristic approaches can be used to gauge patterns of inter-locus heterogeneity in the absence of detailed parameter estimation. Plotting allele frequencies at individual loci against average frequencies taken across loci can point to genomic regions with unusual behaviour (Szymura & Barton 1986). Additionally, visually inspecting the distribution of parameter values across loci is a helpful strategy. When enough loci have been surveyed, the existence of outliers or multiple modes could reveal differential introgression.

Genomic clines

Geographic clines connect the distances between populations in space with the gradient in allele frequencies. Because the highest proportion of hybrids occurs in the centre of the hybrid zone, geographic distance can be viewed as a surrogate measure of the degree of hybridization. An alternative approach is to directly use hybrid ancestries across the genome to predict introgression at individual loci.

Consider a scenario in which multiple, unlinked loci are surveyed in individuals sampled from a hybrid zone. Collectively, genotypes at these loci measure the degree of admixture for each individual. In particular, if allele frequencies in source populations are available, the fraction of the genome derived from each population can be estimated. This 'hybrid index' (h) ranges from 0 (all alleles inherited from one parental population) to 1 (all alleles inherited from the other parental population). For a set of completely informative loci, where alternative alleles are fixed in the source populations and inherited codominantly, the hybrid index is simply the fraction of alleles from each population (Barton & Gale 1993). More sophisticated approaches based on likelihood (Buerkle 2005) and Bayesian clustering (Pritchard *et al.* 2000; Falush *et al.* 2003; Durand *et al.* 2009) allow estimation of hybrid indices from loci that show polymorphism or dominance within source populations. Calculations are performed for each individual in the hybrid zone, yielding a collection of hybrid ancestries.

The distribution of hybrid indices provides a great deal of information about the forces that maintain the structure of a hybrid zone (Harrison & Bogdanowicz 1997; Jiggins & Mallet 2000). For example, a bimodal distribution (a relative paucity of F_1 genotypes) suggests the presence of prezygotic barriers to mating or fertilization, while a unimodal distribution points to other forces (Jiggins & Mallet 2000). The hybrid index distribution can also be used to generate the pattern of introgression expected in the absence of selection (Lexer *et al.* 2007; Gompert & Buerkle 2009). Genomic cline analysis uses this strategy. My description of the method closely follows Gompert & Buerkle (2009).

With two allelic classes, there are three possible genotypic categories. For the purpose of describing the approach, denote each genotypic category (Y) by the number of alleles it has from a particular source population: 0, 1 or 2. The relative probabilities of each genotype can be related to the genome-wide hybrid index by treating the $Y = 0$ class as a reference and using logit functions of the form

$$\begin{aligned} g_1(h) &= \ln\left(\frac{P(Y=1|h)}{P(Y=0|h)}\right) = \alpha_1 + \beta_1 h \\ g_2(h) &= \ln\left(\frac{P(Y=2|h)}{P(Y=0|h)}\right) = \alpha_2 + \beta_2 h \end{aligned} \tag{9}$$

.

The conditional probabilities (expected frequencies) of each genotype given the hybrid index are then

$$\begin{aligned} \pi_0 &= P(Y=0|h) = \frac{1}{1 + e^{g1(h)} + e^{g2(h)}} \\ \pi_1 &= P(Y=1|h) = \frac{e^{g1(h)}}{1 + e^{g1(h)} + e^{g2(h)}} \\ \pi_2 &= P(Y=2|h) = \frac{e^{g2(h)}}{1 + e^{g1(h)} + e^{g2(h)}} \end{aligned} \tag{10}$$

Assuming a multinomial distribution of genotypes in the population, the log-likelihood of the observed genotype counts across all individuals is

$$\ln L(n_0, n_1, n_2) = n_0 \ln \pi_0 + n_1 \ln \pi_1 + n_2 \ln \pi_2 \tag{11}$$

This expression measures the likelihood of observing an array of genotype counts at any locus in the genome given the distribution of individual hybrid indices. To detect locus-specific selection, this likelihood needs to be compared to the likelihood of observing the data under a neutral model. Gompert & Buerkle (2009) present two alternative procedures for formulating the neutral model.

First, genotypes are randomly permuted among loci within individuals. Multinomial regression is used to estimate the probability of observing each genotype for each permutation, given the hybrid index. The mean probability of observing each genotype in each individual across *all loci* is estimated by repeating this permutation procedure a large number of times. The log-likelihood of the observed genotype counts at a particular locus under the model of neutral introgression is then calculated from the multinomial distribution (as above). The best fitting model at that locus is compared to the neutral model using a likelihood ratio test. The logic of this approach is that under neutrality, individual genotypes at a particular locus simply represent random draws from genomes with the observed collection of hybrid ancestries (h).

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The practice of permuting genotypes across loci to formulate the null model assumes that loci are exchangeable. If loci differ in source population allele frequencies, the exchangeability assumption will be violated. In such cases, an alternative approach that incorporates allele frequency information from outside the hybrid zone can be used (Lexer *et al.* 2007; Gompert & Buerkle 2009). Define the difference between observed and expected heterozygosity in an individual as

$$\Delta_H = H_{\text{observed}} - 2E(f_A)(1 - E(f_A)) \quad (12)$$

f_A as the frequency of the allele from one source population (the *A* allele) and f_{AA}, f_{Aa} and f_{aa} as the frequencies of the three genotypes in the hybrid zone. The expected genotypic frequencies under the neutral model are

$$\begin{aligned} E(f_{AA}) &= E(f_A)^2 - (\Delta_H \times f_{AA}) \\ E(f_{Aa}) &= 2E(f_A)(1 - E(f_A)) + \Delta_H \\ E(f_{aa}) &= (1 - E(f_A))^2 - (\Delta_H \times f_{aa}) \end{aligned} \quad (13)$$

These expected frequencies are used to simulate a large population (where sampling error can be neglected) that has the observed individual hybrid index and heterozygosity values. Multinomial regression again supplies the probabilities of observing each genotype as a function of the hybrid index, and the likelihood of obtaining the actual genotype counts under the model of neutral introgression can be calculated as above. A likelihood ratio test can be used to compare this null model to the best fitting model at each locus.

To estimate critical values for the likelihood ratio tests in both approaches, Gompert & Buerkle (2009) again use simulations to create a distribution of likelihood ratio statistics that would be expected if all observed deviations from neutral predictions were solely attributable to sampling error.

Genomic clines can also inform on the *mode* of selection when loci depart from neutral introgression. For example, loci experiencing underdominant selection will show a deficit of heterozygotes across the scale of hybrid indices, but especially in the individuals who are highly admixed. Epistatic selection against hybrid two-locus combinations produces shifts that are similar to but less dramatic than those seen with underdominance.

Comparing introgression by genomic location

An especially useful strategy for comparing introgression among loci is to organize surveyed markers by their genomic locations. For some species, the genomic compartments where markers reside – autosomal, sex-chromosomal, mitochondrial or chloroplast – are the only available positional information. In these cases, knowledge of the genetics of speciation, as well as likely effects of differences in inheritance on introgression, motivates the comparison among loci in different compartments. Alternatively, when marker positions are known from genetic or physical maps, the spatial organization of clinal patterns can be examined. This approach is particularly powerful when multiple loci from the same chromosome have been surveyed (Payseur *et al.* 2004; Carling & Brumfield 2009). For example, Carling & Brumfield (2009) surveyed 10 *Z*-linked markers in a hybrid zone between *Passerina amoena* and *Passerina cyanea* buntings. Geographic clines' analysis revealed a striking pattern: one locus (VLDLR9) does not flow through the hybrid zone despite clear introgression at the other nine markers (Figs 1 and 2). The positions of flanking markers on the Z chromosome of zebra finch (marker positions are unknown in bunting) suggest that the distal boundary of the region of reduced introgression might lie near the marker with reduced cline width, while the boundary on the proximal side could extend for several megabases (Fig. 2). One or more genes that confer a reproductive barrier between bunting species might reside in this genomic region.

The existence of detailed genetic and physical maps for hybridizing taxa allows genomic sources of variation in introgression to be identified. Whether gene flow across hybrid zones is associated with local recombination rate is worth testing for two reasons. First, multiple models predict that mutations causing reproductive

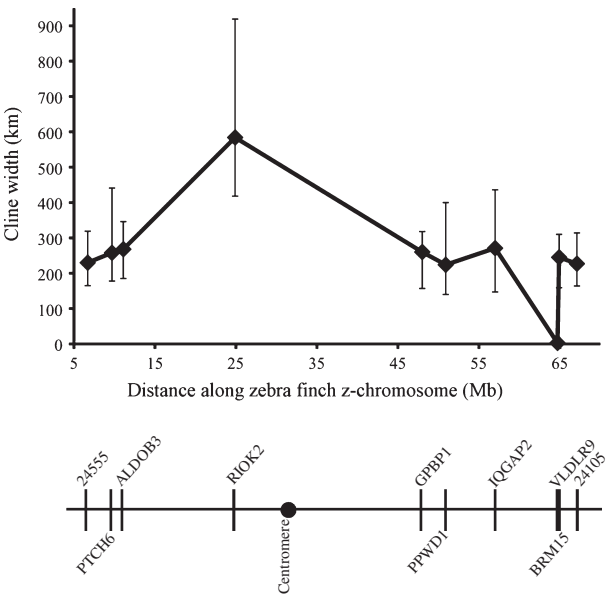


Fig. 2 Variation in geographic cline width across the Z chromosome in a hybrid zone between *Passerina amoena* and *Passerina cyanea*. Cline widths were estimated using maximum likelihood, and error bars indicate 2-unit likelihood support limits. Reproduced and modified from Fig. 3 in Carling & Brumfield (2009).

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isolation can accumulate in genomic regions with restricted recombination (Noor *et al.* 2001; Rieseberg 2001; Butlin 2005; Kirkpatrick & Barton 2006). Second, a general correlation between rates of introgression and recombination could arise if many loci contribute to reproductive isolation (Payseur & Nachman 2005). When genome sequences are available, introgression can be compared to other genomic variables of interest (Payseur & Nachman 2005). Most importantly, candidate genes for reproductive barriers can be nominated based on the positions of regions showing reduced introgression.

Linkage disequilibrium

Patterns of introgression at individual loci are also shaped by processes occurring elsewhere in the genome. When the recombination fraction between two loci is similar to or less than the strength of selection, substantial linkage disequilibrium is expected in the centre of the cline (Slatkin 1975). With strong selection, gene flow can generate linkage disequilibrium between unlinked loci (Slatkin 1975).

In general, when significant linkage disequilibrium exists in hybrid zones, it is expected to reflect associations between conspecific alleles. Individuals dispersing into the hybrid zone carry allelic combinations from the parental populations; differing allele frequencies and admixture generate linkage disequilibrium among alleles (Li & Nei 1974). The subsequent decay of linkage disequilibrium with independent assortment and recombination is opposed primarily by three forces. First, migration continues to add parental combinations of alleles to the hybrid zone. Second, positive assortative mating maintains associations among alleles. Third, selection against hybrids removes certain genotypic combinations. In the case of epistatic incompatibilities, selection purges hybrid combinations of alleles at multiple loci, leaving behind associations between conspecific alleles (Gardner *et al.* 2002; Payseur & Hoekstra 2005). Interestingly, dispersal can be a stronger determinant of linkage disequilibrium than epistatic selection when a large number of loci contribute to fitness (Slatkin 1975; Barton 1983).

Pairwise linkage disequilibrium in hybrid zones can be measured using standard approaches. Iterative algorithms such as expectation-maximization (EM) can be used to estimate two-locus gametic frequencies, from which D' (Lewontin 1964), R^2 (Hill & Robertson 1968) and other measures of linkage disequilibrium can be calculated. EM methods for estimating two-locus gametic frequencies typically require Hardy–Weinberg equilibrium, a balance that is often violated in hybrid zones; methods that apply directly to observed two-locus genotypes (Weir 1996; Weir *et al.* 2004) might be more useful. The average amount of linkage disequilibrium can also be estimated indirectly by calculating the inter-individual variance in the hybrid index (Barton & Gale 1993). This variance rises when alleles that increase the hybrid index are associated.

When the physical locations of surveyed markers are known, the rate of decay of linkage disequilibrium with distance along the chromosome provides additional information about introgression. Hybrid chromosomes are composed of blocks with alternating ancestries. Alleles within blocks are derived from the same parental population, whereas alleles from neighbouring blocks are derived from different parental populations. Each switchpoint or junction represents a recombination event during the history of the hybrid zone, and the number of junctions increases with more generations of hybridization (Fisher 1953; Baird 2006). Hence, the distribution of block lengths can be used to estimate the age of a hybrid zone (Baird 1995) as well as the migration rate (Pool & Nielsen 2009). Genomic regions that fail to introgress owing to selection against hybrids might have relatively longer block lengths. This logic raises the possibility of detecting differential introgression from the distribution of block lengths when sufficient marker density is available. Such an approach would be most applicable in species for which recombination rate estimates are available.

Using introgression at randomly chosen loci to find the targets of selection in hybrid zones

When the locus experiencing selection in hybrids has been surveyed, inferences about selection can be drawn based on observed patterns of allele and genotype frequencies. However, empirical studies of hybrid zones will rarely sample selected loci directly without prior knowledge. Instead, most surveyed loci serve as markers for introgression of particular genomic segments. How are these randomly chosen loci expected to reflect selection that targets other loci?

The effect of selection acting elsewhere in the genome on the barrier to gene flow at neutral loci has been evaluated using a variety of theoretical models (Bazykin 1969, 1972a,b; Karlin & Mcgregor 1972; Barton 1979, 1983; Spirito *et al.* 1983; Bengtsson 1985; Barton & Bengtsson 1986; Barton & Gale 1993; Baird 1995; Gavrilets 1997, 2004; Gavrilets & Cruzan 1998). A conceptually simple result is that selection acting against heterozygotes (hybrids) at one locus with strength s will engender a barrier to gene flow (b) of

$$b = \frac{1+s}{1-s} \tag{14}$$

at a neutral locus unlinked to the selected gene, and a barrier of

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$$b = \frac{s}{r(1-s)} \quad (15)$$

at a neutral locus closely linked to the selected gene, where r is the recombination fraction between the neutral and selected loci (Bengtsson 1985). b can be interpreted as the number of subpopulations that would need to be inserted in the hybrid zone array to create a spatial barrier to gene flow equivalent to that formed by selection (and is the discrete-space analogue of the barrier for populations in continuous space B, above). Hence, the barrier to gene flow at a neutral locus increases with the strength of selection against hybrids and with tighter linkage to the target of selection.

The general principle that the level of introgression at a randomly chosen locus depends largely on the relative magnitudes of selection and recombination also characterizes results from more realistic models in which selection acts on multiple loci. The results from one study (Gavrilets 1997) are worth highlighting because of their relevance to the dynamics of gene flow in real hybrid zones. Gavrilets (1997) considered a three-locus, two-allele model in which fitness is specified by the interaction between two loci (*A* and *B*) and a third locus represents a neutral marker. The conspecific diploid genotypes *AABB* and *aabb* represent peaks in the adaptive landscape, with the remaining seven genotypes experiencing epistatic selection of potentially differing strengths. Compelling evidence for the involvement of this type of 'Dobzhanksy-Muller incompatibility' in postzygotic isolation across a wide variety of species (Coyne & Orr 2004) makes this framework useful. Gavrilets (1997) used a stepping-stone model of migration to derive analytical approximations for equilibrium allele frequencies at selected and marker loci in a hybrid zone. This model demonstrates that, like the case of heterozygote disadvantage, marker allele frequency clines are distorted by strong epistatic selection on other loci; the magnitude of the effect depends on the migration rate. Frequent migration into the centre of the hybrid zone largely erases the effects of selection on marker loci. But when migration is rare, selection against F_1 hybrids increases linkage disequilibrium, causing clines for neutral alleles to resemble those at selected loci. Selection must be strong for this effect to be observed – even a small amount of recombination (e.g. $r = 0.05$; Gavrilets 1997) can uncouple clines at linked markers and incompatibilities if hybrid fitness is not substantially reduced. Results also predict that clines at markers linked to strong incompatibilities should be disjointed and asymmetrical, whereas loci unlinked to selected targets will generally show concordant clines (Barton 1983).

Gavrilets (1997) focused on hybrid inviability, which is only one component of reproductive isolation between species. Interestingly, one-locus models indicate that assortative mating and fertility selection have the potential to impede the introgression of neutral alleles more effectively than postzygotic isolation (Gavrilets & Cruzan 1998).

Confounding factors in interpreting inter-locus heterogeneity in introgression

The basic premise of comparative studies of introgression is that loci with reduced gene flow relative to the remainder of the genome point to regions that may be involved in reproductive isolation. If homogeneous selection and migration were the only persistent forces operating in hybrid zones, perhaps we could be confident in this interpretation. Unfortunately, additional factors are expected to increase inter-locus heterogeneity in gene flow.

Although different loci experience the same migration rate, genetic drift generates inter-locus variance in allele frequencies when loci are unlinked. This effect creates differential introgression in the absence of selection, especially in small populations. The rate of drift also determines the degree to which the phylogenetic history of individual loci matches the population history. In species with recent divergence times or large effective population sizes (or both), incomplete sorting of ancestral polymorphism produces a complex mixture of genealogies across the genome (Hudson & Coyne 2002; Broughton & Harrison 2003; Machado & Hey 2003; Pollard *et al.* 2006; White *et al.* 2009), substantially increasing inter-locus heterogeneity in patterns of variation. Additionally, incomplete lineage sorting and recent gene flow leave similar footprints (shared variation) at individual loci, raising the possibility of confusing the two processes in hybrid zones. Although the common practice of using markers that are diagnostic for species should reduce this confounding, incomplete lineage sorting nevertheless provides an important source of inter-locus variance in hybrid zones.

The inheritance process can contribute additional inter-locus heterogeneity to patterns of introgression. Variants in different genomic compartments (mitochondrial DNA, chloroplast DNA, sex chromosomes and autosomes) record distinct ancestries. Two aspects of this phenomenon bear directly on interpretations of gene flow in hybrid zones. First, variation in effective population size leads to disparate rates of genetic drift. For example, allele frequencies will diverge faster by chance at Y-linked loci than autosomal loci. Second, genomic compartments are expected to show discordant levels of introgression in species with sex-biased dispersal. Differences in gene flow among loci with differing modes of inheritance are commonly observed in

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hybrid zones. The interpretation that this inter-locus variance reflects differential selection should be considered with caution.

Other genomic variables contribute to differences in introgression among loci. As outlined earlier, the power to detect loci that confer reproductive barriers between species is influenced by the amount of linkage disequilibrium between these loci and surveyed markers. Because recombination shapes the physical scale of linkage disequilibrium and varies across the genome, loci with identical selective effects can differentially distort introgression of markers in different parts of the genome. The effects of variation in recombination rate can be ameliorated by choosing large numbers of markers that provide improved genomic coverage but continues to be an issue with the marker densities currently employed in hybrid zones. Variation in mutation can also increase inter-locus heterogeneity in introgression, particularly when variants mutate according to different rates and mechanisms (e.g. microsatellites and SNPs) (Yuri *et al.* 2009). Fortunately, these effects are only likely to be significant when mutation rates exceed migration rates, an uncommon scenario in hybrid zones.

Along with migration, selection and sampling error, these factors collectively shape the portrait of gene flow across the genome that is revealed by molecular markers. Similar complications challenge attempts to identify loci experiencing selection as empirical outliers in genomic patterns of variation within species. In this context, some lessons have been learned that are likely to apply to the study of hybrid zones. Inaccurate demographic assumptions routinely produce false-positive signals of selection in outlier approaches (Kelley *et al.* 2006; Thornton & Jensen 2007; Thornton *et al.* 2007; Excoffier *et al.* 2009). Additionally, selected loci need not lie in the most extreme parts of the genomic distribution of variation (Teshima *et al.* 2006). As genomic comparisons of introgression begin to accumulate, the expected performance of outlier approaches in hybrid zones is an important area for theoretical attention.

Emerging empirical patterns

Hybrid zone studies that compare patterns of introgression at 10 or more loci are listed in Table 1. Although these studies use a variety of taxa (with a current bias towards sunflowers and house mice), sampling designs and analytical methods, several consistent trends emerge. The repeated observation of substantial variance in introgression across the genome is the most notable pattern. Some studies report a few regions showing little introgression in the face of rampant gene flow elsewhere in the genome (Carling & Brumfield 2009), while others describe a more uniform distribution of clines among loci (Teeter *et al.* 2008). That differential introgression appears to be a common theme in hybrid zones suggests caution in drawing conclusions based on a few loci and magnifies the prospects of identifying genomic regions that confer reproductive barriers between species. It is noteworthy that reduced introgression is detectable with relatively low marker densities, raising two potential interpretations. First, small numbers of anonymously chosen markers effectively tag selection occurring at other loci, reflecting a slow decay of linkage disequilibrium or the existence of many genes that contribute to reproductive isolation. Second, those loci that show reduced introgression do not mark selection against mixing in particular genomic regions but instead represent the outcomes of mostly neutral evolutionary processes. Distinguishing between these competing explanations is an important challenge for future studies of hybrid zones.

A handful of studies in Table 1 compare introgression at the same loci in multiple, independent hybrid zone transects. This is an especially powerful approach because loci that cause (intrinsic genetic) reproductive isolation should show reduced introgression consistently, whereas neutral regions are unlikely to lie in the tail of the genomic distribution in multiple transects. Interestingly, the same loci either correspond (Buerkle & Rieseberg 2001) or diverge (Nolte *et al.* 2009; Teeter *et al.* 2010) in their patterns of introgression in different transects, depending on the study. Concordance among transects could be limited by the genetic complexity of reproductive isolation (Teeter *et al.* 2010). For example, intraspecific polymorphism at loci that cause hybrid sterility and hybrid inviability could generate discordance in levels of gene flow in different transects. The analytical challenges reviewed here also contribute to this heterogeneity.

Some studies in Table 1 identify biases in the genomic location of markers showing reduced gene flow. For example, the X chromosome exhibits lower introgression than the autosomes in multiple transects of a hybrid zone between species of house mice (Macholan *et al.* 2007; Teeter *et al.* 2008, 2010), confirming earlier patterns based on a small number of markers (Tucker *et al.* 1992; Dod *et al.* 1993). This result might reflect a role for the X chromosome in hybrid male sterility, which is supported by genetic mapping in the laboratory (Oka *et al.* 2004; Storchova *et al.* 2004; Good *et al.* 2008a,b). As predicted by the general significance of the sex chromosomes in reproductive isolation (Coyne & Orr 2004), reduced sex-linked introgression is also observed in other species (Hagen & Scriber 1989; Sperling & Spence 1991; Porter *et al.* 1997; Saetre *et al.* 2003; Carling & Brumfield 2008).

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Challenges and prospects for genomic studies of introgression in hybrid zones

Advantages and limitations of existing analytical approaches

Existing approaches for measuring differential introgression in hybrid zones offer several benefits. The geographic clines framework rests on rich theoretical foundations that tie spatial patterns in allele frequencies to underlying evolutionary forces. As a result, cline shapes provide detailed information about the geographic scales and intensities of selection and migration that would be difficult to extract using other approaches. Predicting introgression at individual loci from the degree of hybrid ancestry – the genomic clines method – offers additional advantages. The nonselective forces that distort single-locus clines should also affect hybrid indices, raising the possibility that the genomic clines approach takes these factors into account. The convenient and logical regression-based framework of the genomic clines method allows differential introgression to be clearly identified and the underlying form of selection to be inferred. Because additional loci will increase the accuracy of hybrid ancestry estimates, method performance should improve with expanding genomic coverage. The magnitude and scale of linkage disequilibrium in hybrid zones provide additional details about the dynamics of migration and selection and indicate how much of the genome is effectively tagged by a particular set of molecular markers.

Challenges and limitations also accompany existing analytical methods. Sensitivities to several departures from model assumptions have yet to be examined. Geographic clines analysis typically assumes that allele frequencies change monotonically across transects (but see (Marshall & Sites 2001; Macholan *et al.* 2007)). In addition to motivating the reduction in a complicated biological scenario to one dimension, this requirement of ordered allele frequencies effectively ignores hybrid zones that are geographically patchy (Harrison and Rand 1989). More generally, the optimal spatial scale for hybrid zone analysis is unclear, and results might differ across scales (Patton 1993; Sites *et al.* 1995). Although the genomic clines method nicely incorporates the effects of genetic drift, false positives can arise when drift occurs independently across sampling localities (Gompert & Buerkle 2009). Whether different forms of population structure can also elevate the rejection rate of the hypothesis of neutral introgression remains to be tested (Teeter *et al.* 2010).

Another potential complication concerns marker selection. Using fixed differences between species as markers of introgression in hybrid zones – a widespread practice – can bias investigators towards inferences of lower introgression (Sattler & Braun 2000; Brumfield *et al.* 2001; Yatabe *et al.* 2007; Yuri *et al.* 2009). Paradoxically, including more genomic regions that contain genes causing reproductive isolation, which will probably harbour markers with fixed differences, will make it increasingly difficult to detect reduced introgression in these regions through comparison to other loci.

The goal of using differential introgression to identify loci involved in reproductive isolation is further hampered by the simultaneous operation of multiple forms of selection in hybrid zones. For example, when compared with loci undergoing adaptive introgression, a higher fraction of the genome will appear to experience reduced gene flow.

An especially difficult challenge for existing analytical methods is that the expected genomic distribution of introgression remains unclear for many realistic biological scenarios. To envision the complexity of this problem, consider a hypothetical hybrid zone in which migration approximates a stepping-stone model, population sizes (and rates of genetic drift) vary among localities and recombination rates differ across the genome. Twenty genes with different phenotypic effects (and selection coefficients) contribute to reproductive isolation between species. Introgression is surveyed at 50 randomly chosen SNPs at variable distances from the 20 genes. What should the distribution of clinal statistics at these 50 SNPs look like? The answer is difficult to come by, and real hybrid zones are more complex than in this scenario.

The complicated nature of patterns of differential introgression should motivate simulation-based comparisons of method performance across a range of demographic scenarios. Understanding the extent to which departures from migration-selection-drift equilibrium in hybrid zones can create differential introgression would be especially helpful. The realistic possibility that source populations also deviate from equilibrium (e.g. because of range expansions) should be considered.

Promising new directions

As in other areas of evolutionary genetics (Gilad *et al.* 2009; Rokas & Abbot 2009), advances in genotyping and sequencing technologies have the potential to revolutionize hybrid zone studies. Molecular markers are no longer in short supply, even in species without advanced genomic tools. The increasing capacity and affordability of next-generation sequencing portends an exciting achievement in the near future: the description of genome sequences in animals and plants from hybrid zones. The prospects for bringing data collection in hybrid zones to

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the genomic scale should motivate careful thinking about analytical approaches.

As the number of surveyed loci grows, methods that better incorporate inter-locus associations will become especially appealing for several reasons. First, as outlined earlier, hybrid zone theory connects linkage disequilibrium to processes of interest, including migration and selection. Second, surveys of introgression with many markers suggest that nonrandom associations among loci are pervasive in hybrid zones, even when loci map to different chromosomes (Teeter *et al.* 2008, 2010). Most importantly, the prevailing explanation for hybrid sterility and hybrid inviability – the Dobzhansky-Muller model – posits that reproductive isolation is caused by disrupted interactions among loci (Bateson 1909; Dobzhansky 1937; Muller 1942; Coyne & Orr 2004). In this epistatic framework, the specific array of multi-locus combinations observed in hybrid zones should reveal important clues about the loci underlying reproductive barriers. Interestingly, initial applications of the genomic clines approach, which can differentiate among forms of selection, suggest limited evidence for epistatic selection (Nolte *et al.* 2009; Teeter *et al.* 2010). Instead, many loci show patterns consistent with selection acting against single-locus heterozygotes (underdominance). Although the bias towards underdominance could reflect the existence of undetected chromosomal rearrangements, it is surprising because selection against heterozygotes should act to prevent fixation of these mutations within species. Much of the theory connecting geographic clines to reproductive barriers has also focused on underdominance at single or multiple loci (Bazykin 1969; Barton 1979, 1983, 2001; Bengtsson 1985; Barton & Bengtsson 1986; Barton & Gale 1993). Despite the likely importance of epistatic selection in shaping genetic variation in hybrid zones, existing analytical approaches are not specifically designed to find it. Epistatic selection does not always cause strong distortions in genotypic frequencies (Gompert & Buerkle 2009), suggesting that new analytical frameworks will be required to meet this salient challenge.

Another analytical strategy that will be enabled by increases in genomic coverage is modelling the positions and sizes of chromosomal blocks inherited from one parental species or the other. Perhaps regions of low introgression could be identified as blocks that are unusually long compared to the genomic distribution. Additionally, the distribution of block lengths could be used to calibrate realistic null models for geographic and genomic clines. Block length distribution is a sensitive indicator of historical changes in migration (Pool & Nielsen 2009). Examination of the ways in which selection against hybrids affects block length would be a fruitful direction for future theoretical studies.

A major advantage of using differential introgression to identify genomic regions involved in speciation is that the phenotypes experiencing selection in hybrids need not be known. Nevertheless, understanding the mechanisms that cause differential introgression in hybrid zones will ultimately require study of the specific traits that reproductively isolate the parental species. Jointly examining the genetics of reproductive barrier phenotypes through crosses in the laboratory and the flow of different genomic regions in natural hybrid zones is an especially powerful approach (Rieseberg *et al.* 1999). If the trait being investigated in the laboratory confers a strong reproductive barrier in nature, genomic regions that affect this trait should resist introgression in hybrid zones. Conversely, the types of loci showing reduced introgression could nominate phenotypes for further study in the laboratory. Ideally, the integration of phenotype and genotype will eventually happen within the context of hybrid zones. Indeed, the identification of genomic regions that fail to introgress should motivate the measurement of reproductive barriers in natural hybrids, already a tradition in many classic hybrid zones. Consideration of the phenotype will be needed for distinguishing between endogenous and exogenous forms of selection against hybrids, a challenging task using clinal patterns alone (Kruuk *et al.* 1999).

As surveys of differential introgression accumulate, it will be useful to compare results across species pairs at different stages of speciation. One prediction is that the number of genomic regions at which gene flow is blocked should increase with divergence time. These studies would provide interesting parallels to comparative analyses of reproductive isolation phenotypes (Coyne & Orr 1989, 1997; Sasa *et al.* 1998; Presgraves 2002; Price & Bouvier 2002; Fitzpatrick 2004), which have yielded important insights into the dynamics of speciation.

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